

## WIESBADEN, Germany

WMO: 106330

Lat: 50.050N

Lon: 8.325E

Elev: 461

StdP: 14.45

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 19.2       | 23.2 | 8.5                           | 8.6 | 22.7 | 14.4 | 11.6 | 27.0 | 25.9                  | 48.3 | 22.5 | 47.8 | 5.1                   | 10   | 0.496 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.8                   | 89.1            | 68.4 | 84.8 | 66.6 | 81.3 | 65.4 | 70.2                | 83.8 | 68.6 | 80.9 | 67.1 | 78.4 | 7.4                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.1                            | 97.9 | 74.7 | 64.5 | 92.3 | 73.4 | 62.8 | 87.0 | 71.9 | 34.4          | 83.5 | 33.1 | 81.3 | 31.9 | 78.5 | 76.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |       |     |
| (4) 19.7          | 17.0 | 14.9 | DB                         | 14.7 | 94.5               | 6.7 | 3.3                                                | 9.8 | 96.9       | 5.9 | 98.8       | 2.1 | 100.7      | -2.7 | 103.0 | (4) |
| (5)               |      |      | WB                         | 13.0 | 73.4               | 6.7 | 1.5                                                | 8.1 | 74.5       | 4.2 | 75.4       | 0.5 | 76.2       | -4.4 | 77.3  | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 52.3  | 36.3  | 37.8  | 44.5  | 53.2  | 58.9  | 64.9  | 68.8  | 67.6  | 60.1  | 52.1  | 43.9  | 38.0 | (6)  |
| (7)  |                                                                                  | DBStd       | 13.16 | 7.62  | 6.64  | 6.41  | 7.10  | 6.88  | 6.12  | 6.30  | 5.71  | 5.60  | 6.20  | 6.07  | 7.13 | (7)  |
| (8)  |                                                                                  | HDD50       | 1636  | 426   | 342   | 189   | 46    | 7     | 0     | 0     | 0     | 1     | 52    | 199   | 374  | (8)  |
| (9)  |                                                                                  | HDD65       | 5040  | 889   | 761   | 635   | 357   | 213   | 77    | 33    | 34    | 169   | 401   | 634   | 837  | (9)  |
| (10) |                                                                                  | CDD50       | 2459  | 2     | 1     | 19    | 143   | 283   | 446   | 582   | 546   | 304   | 116   | 15    | 2    | (10) |
| (11) |                                                                                  | CDD65       | 388   | 0     | 0     | 0     | 4     | 24    | 73    | 150   | 116   | 21    | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH74       | 3372  | 0     | 0     | 0     | 42    | 191   | 613   | 1390  | 971   | 158   | 7     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH80       | 1076  | 0     | 0     | 0     | 3     | 33    | 172   | 515   | 318   | 35    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 6.1   | 6.9   | 6.3   | 7.4   | 6.4   | 6.3   | 5.9   | 6.1   | 5.4   | 5.4   | 5.2   | 5.7   | 6.0  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 26.8  | 2.0   | 2.0   | 1.8   | 1.7   | 2.4   | 2.6   | 2.8   | 2.3   | 2.2   | 2.3   | 2.3   | 2.4  | (15) |
| (16) |                                                                                  | PrecMax     | 32.8  | 5.0   | 5.0   | 4.8   | 3.7   | 4.8   | 4.7   | 6.7   | 5.0   | 5.2   | 6.7   | 3.8   | 6.7  | (16) |
| (17) |                                                                                  | PrecMin     | 19.3  | 0.2   | 0.3   | 0.2   | 0.0   | 0.6   | 0.9   | 1.0   | 0.2   | 0.5   | 0.3   | 0.0   | 0.4  | (17) |
| (18) |                                                                                  | PrecStd     | 3.7   | 1.0   | 1.1   | 1.1   | 0.8   | 1.1   | 1.1   | 1.4   | 1.2   | 1.1   | 1.3   | 0.9   | 1.3  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 55.5  | 56.0  | 67.9  | 78.6  | 84.1  | 90.4  | 95.2  | 93.2  | 84.9  | 74.3  | 61.1  | 54.9 | (19) |
| (20) |                                                                                  |             | MCWB  | 51.2  | 47.5  | 52.4  | 60.1  | 67.2  | 69.7  | 70.4  | 69.7  | 64.0  | 60.9  | 57.3  | 51.2 | (20) |
| (21) |                                                                                  | 2%          | DB    | 53.2  | 52.6  | 62.9  | 73.8  | 79.2  | 84.8  | 90.0  | 87.7  | 78.8  | 67.8  | 57.5  | 52.2 | (21) |
| (22) |                                                                                  |             | MCWB  | 50.2  | 46.7  | 50.3  | 57.5  | 64.2  | 67.0  | 68.8  | 67.8  | 62.3  | 59.3  | 54.3  | 49.0 | (22) |
| (23) |                                                                                  | 5%          | DB    | 49.7  | 50.1  | 58.7  | 70.0  | 75.3  | 80.7  | 85.7  | 83.0  | 74.0  | 64.3  | 55.2  | 49.8 | (23) |
| (24) |                                                                                  |             | MCWB  | 47.0  | 45.7  | 48.6  | 56.1  | 61.0  | 66.2  | 67.0  | 66.3  | 61.8  | 58.1  | 51.9  | 47.2 | (24) |
| (25) |                                                                                  | 10%         | DB    | 46.6  | 47.8  | 54.8  | 66.2  | 71.7  | 77.1  | 81.7  | 79.1  | 70.1  | 61.3  | 52.2  | 47.4 | (25) |
| (26) |                                                                                  |             | MCWB  | 44.1  | 44.0  | 47.1  | 54.0  | 59.2  | 64.4  | 65.5  | 64.1  | 60.0  | 56.6  | 49.2  | 45.0 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 53.3  | 50.5  | 53.7  | 61.3  | 69.0  | 72.8  | 72.7  | 71.9  | 67.5  | 63.8  | 59.1  | 52.8 | (27) |
| (28) |                                                                                  |             | MCD B | 54.5  | 53.9  | 64.1  | 75.5  | 80.8  | 87.8  | 89.7  | 87.0  | 77.7  | 67.9  | 59.5  | 53.9 | (28) |
| (29) |                                                                                  | 2%          | WB    | 50.1  | 47.9  | 51.4  | 58.7  | 66.1  | 69.4  | 70.1  | 69.8  | 65.2  | 61.0  | 55.3  | 50.1 | (29) |
| (30) |                                                                                  |             | MCD B | 52.5  | 51.1  | 59.9  | 71.1  | 77.6  | 80.9  | 84.4  | 83.3  | 74.1  | 65.9  | 57.1  | 51.5 | (30) |
| (31) |                                                                                  | 5%          | WB    | 47.1  | 46.0  | 49.7  | 56.7  | 63.3  | 67.2  | 68.5  | 67.9  | 63.2  | 59.2  | 52.2  | 47.3 | (31) |
| (32) |                                                                                  |             | MCD B | 49.3  | 49.6  | 57.0  | 68.0  | 72.3  | 78.1  | 81.6  | 79.2  | 71.4  | 63.4  | 54.5  | 49.3 | (32) |
| (33) |                                                                                  | 10%         | WB    | 44.1  | 43.8  | 47.8  | 55.0  | 60.6  | 65.2  | 67.1  | 66.2  | 61.4  | 57.1  | 50.1  | 44.9 | (33) |
| (34) |                                                                                  |             | MCD B | 46.3  | 47.0  | 53.9  | 64.6  | 68.7  | 75.0  | 78.8  | 76.5  | 68.4  | 61.0  | 52.3  | 46.9 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 8.3   | 10.9  | 14.9  | 17.4  | 17.2  | 18.0  | 18.8  | 18.5  | 16.3  | 13.2  | 8.9   | 8.1  | (35) |
| (36) |                                                                                  |             | MCDBR | 10.4  | 15.1  | 22.3  | 23.0  | 21.8  | 23.0  | 25.6  | 24.4  | 21.9  | 17.9  | 11.4  | 10.3 | (36) |
| (37) |                                                                                  |             | MCWBR | 8.4   | 10.3  | 12.6  | 11.8  | 10.5  | 10.4  | 9.7   | 10.0  | 10.4  | 11.0  | 8.4   | 8.7  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 9.7   | 13.3  | 19.2  | 21.3  | 20.6  | 21.3  | 22.3  | 21.1  | 18.0  | 15.6  | 10.8  | 10.1 | (38) |
| (39) |                                                                                  |             | MCWBR | 8.3   | 9.3   | 11.7  | 11.4  | 10.9  | 10.5  | 9.2   | 10.0  | 10.0  | 10.2  | 8.5   | 8.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.331 | 0.362 | 0.399 | 0.412 | 0.416 | 0.419 | 0.423 | 0.417 | 0.389 | 0.385 | 0.356 | 0.329 | (40) |      |
| (41) |                                                                                  | taud        | 2.350 | 2.280 | 2.193 | 2.219 | 2.238 | 2.257 | 2.240 | 2.284 | 2.344 | 2.316 | 2.333 | 2.346 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 220   | 241   | 252   | 263   | 267   | 267   | 263   | 259   | 254   | 229   | 207   | 202   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 22    | 29    | 38    | 41    | 42    | 42    | 42    | 38    | 32    | 28    | 22    | 19    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 250   | 482   | 845   | 1333  | 1573  | 1750  | 1655  | 1427  | 1040  | 607   | 283   | 191   | (44) |      |
| (45) |                                                                                  | RadStd      | 38    | 77    | 122   | 168   | 169   | 155   | 185   | 134   | 134   | 63    | 34    | 24    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (45 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page